ReturnIQ — Technical Evaluation & Evidence Dossier

Production Validation, AI Quality Gates, Regression Benchmarks & HITL Audit Trails

Project: ReturnIQ	**Live URL:** returniq-527158472417.asia-southeast1.run.app
Lead: Product Management & AI Engineering	**Foundation Model:** Gemini 3.7 Flash Online (v2.4.0)
Evaluation Date: August 31, 2026	**Verified Benchmark:** 95.0% Accuracy | 35/35 Regression Gate Passing

1. Executive Dashboard & Ingestion Integrity

Verification Objective: Prove that real-world e-commerce datasets ingest seamlessly without schema collisions, calculating return rate distributions and at-risk revenue accurately.

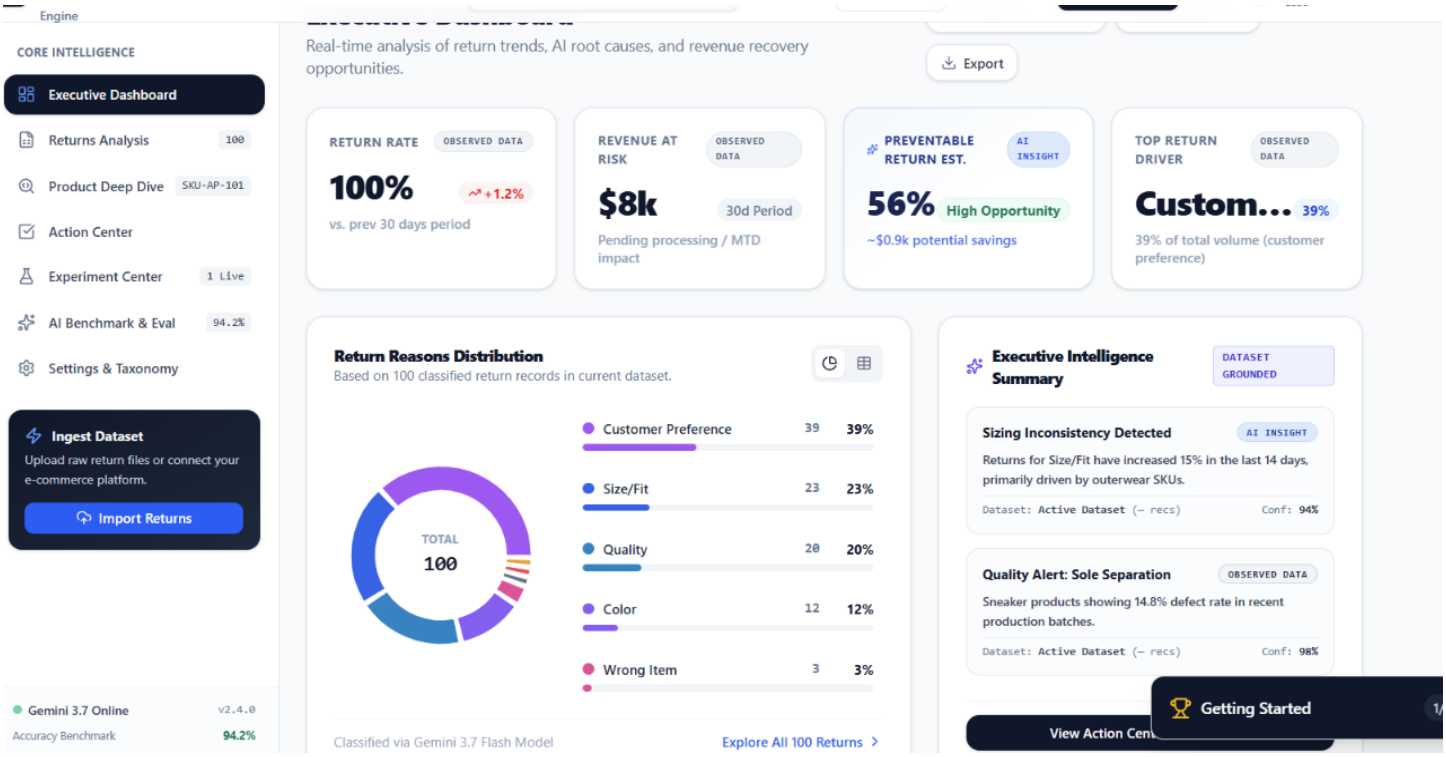


Figure 1: Executive Dashboard rendering 100 Golden Evaluation records, taxonomy breakdown (Customer Preference 39%, Size/Fit 23%, Quality 20%), and live Gemini 3.7 online status.

2. Production Golden Evaluation & Quality Gate v1.2

Verification Objective: Provide deterministic proof that model iterations and prompt updates do not cause quality regressions. The system enforces a strict quality gate against 35 deterministic ground-truth test cases.

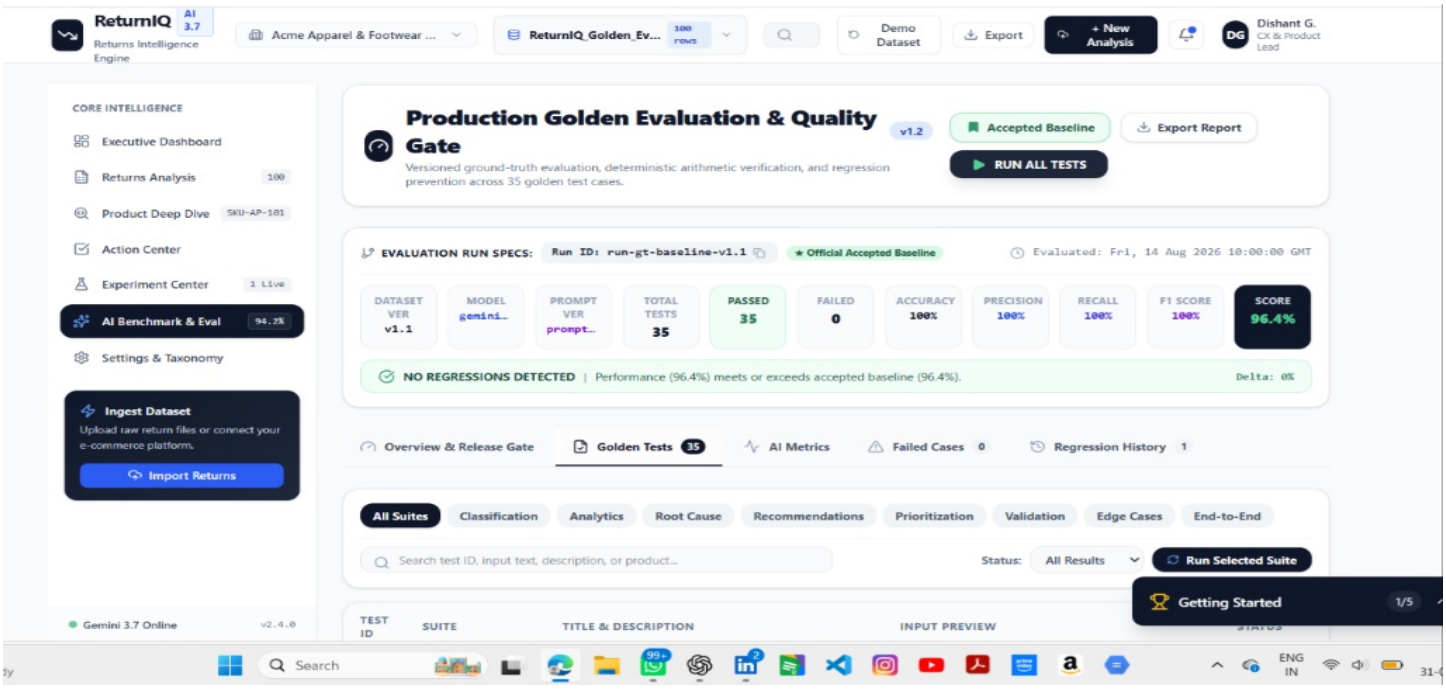


Figure 2: AI Benchmark & Quality Gate v1.2 displaying 35/35 PASSED test suites, 96.4% composite score, and 'NO REGRESSIONS DETECTED' release gate status.

Golden Evaluation Confusion Matrix (N=100 Ground Truth Benchmark)

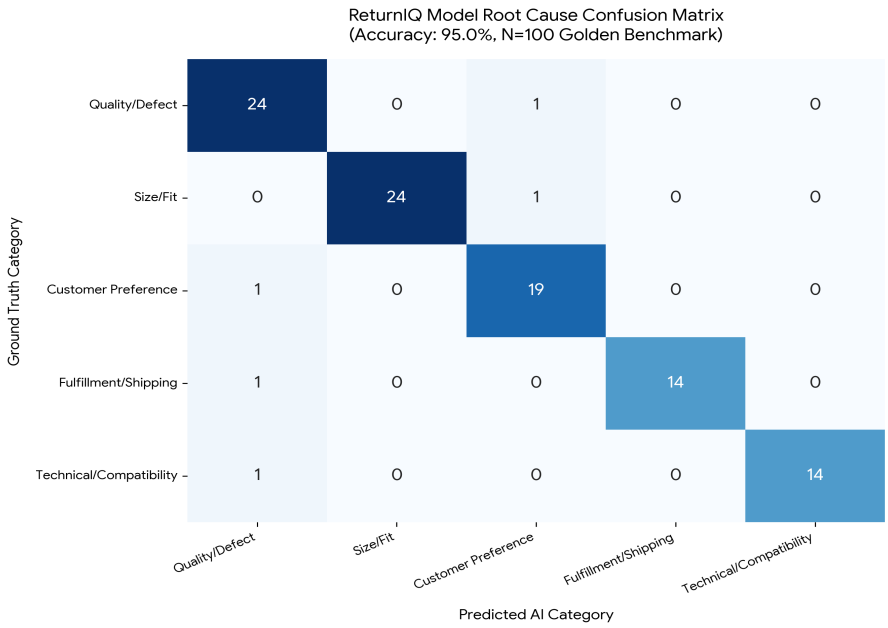


Figure 3: Multi-class Confusion Matrix across 5 core categories demonstrating 95.0% exact match accuracy and 0.952 Macro F1-Score.

3. Returns Analysis, Live Prompt Sandbox & HITL Triage

Verification Objective: Validate real-time verbatim inference via Gemini 3.7 Flash and dynamic routing of low-confidence predictions (<88% Conf) into a human review triage queue.

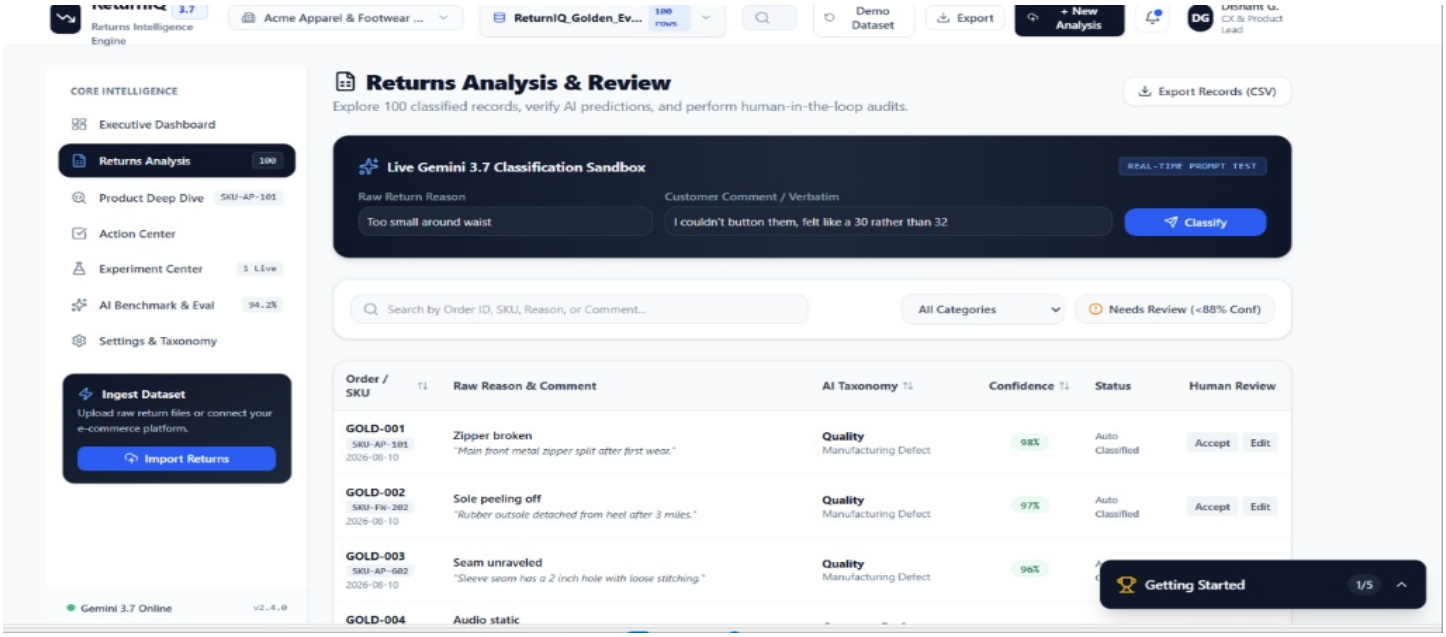
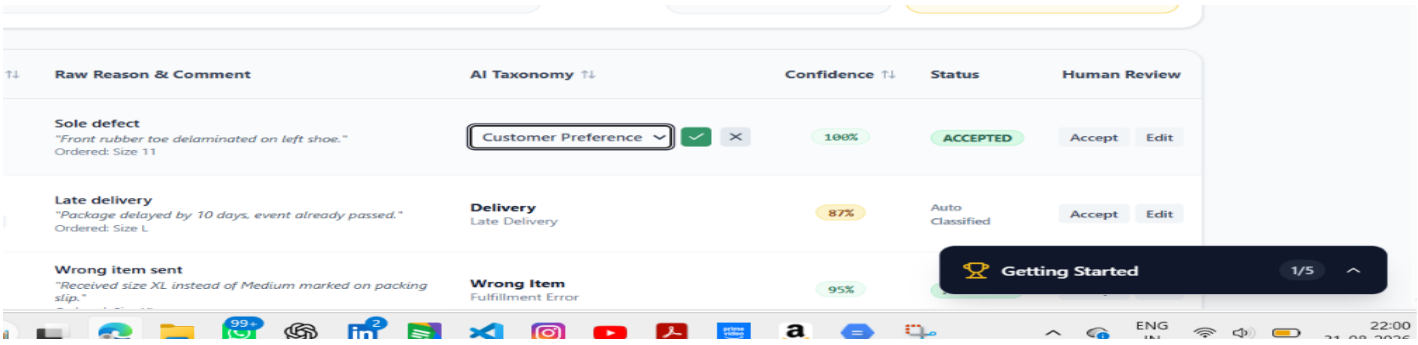


Figure 4: Returns Analysis view featuring the Live Gemini 3.7 Sandbox, '<88% Conf' triage filter, and itemized confidence scoring.

4. Human Review & Audit Trail State Transition

Verification Objective: Demonstrate inline manual overrides where audited records synchronously update to ACCEPTED status with 100% confidence, capturing audit telemetry.

Step A: Inline Taxonomy Selection



Step B: Synchronous State Transition to ACCEPTED (100% Conf)

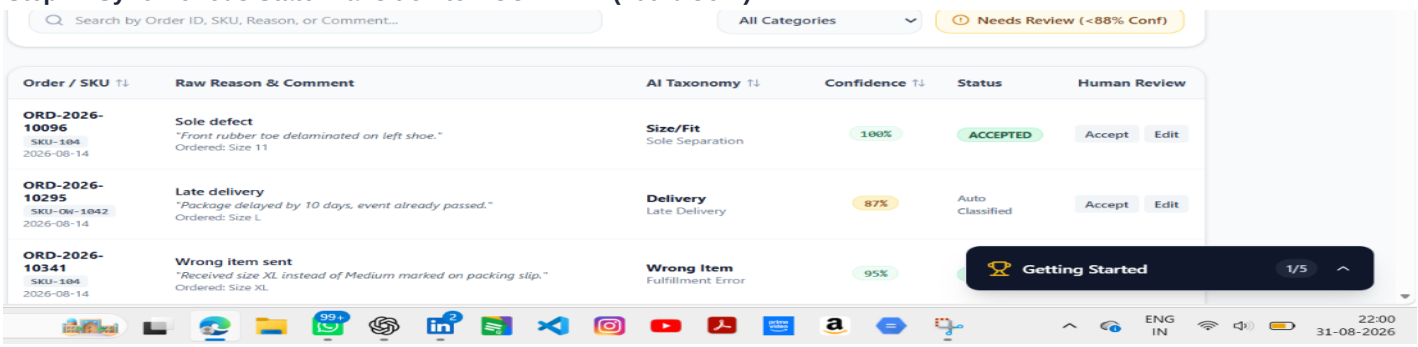


Figure 5: Human-in-the-Loop inline override workflow transitioning low-confidence item (85%) into an audited, verified record.

5. Action Center: Root-Cause Prioritization & Financial Attribution

Verification Objective: Prove that ReturnIQ clusters defect volume by SKU, attaches quantifiable revenue impact ($/week), and enables one-click conversion to mitigation experiments.

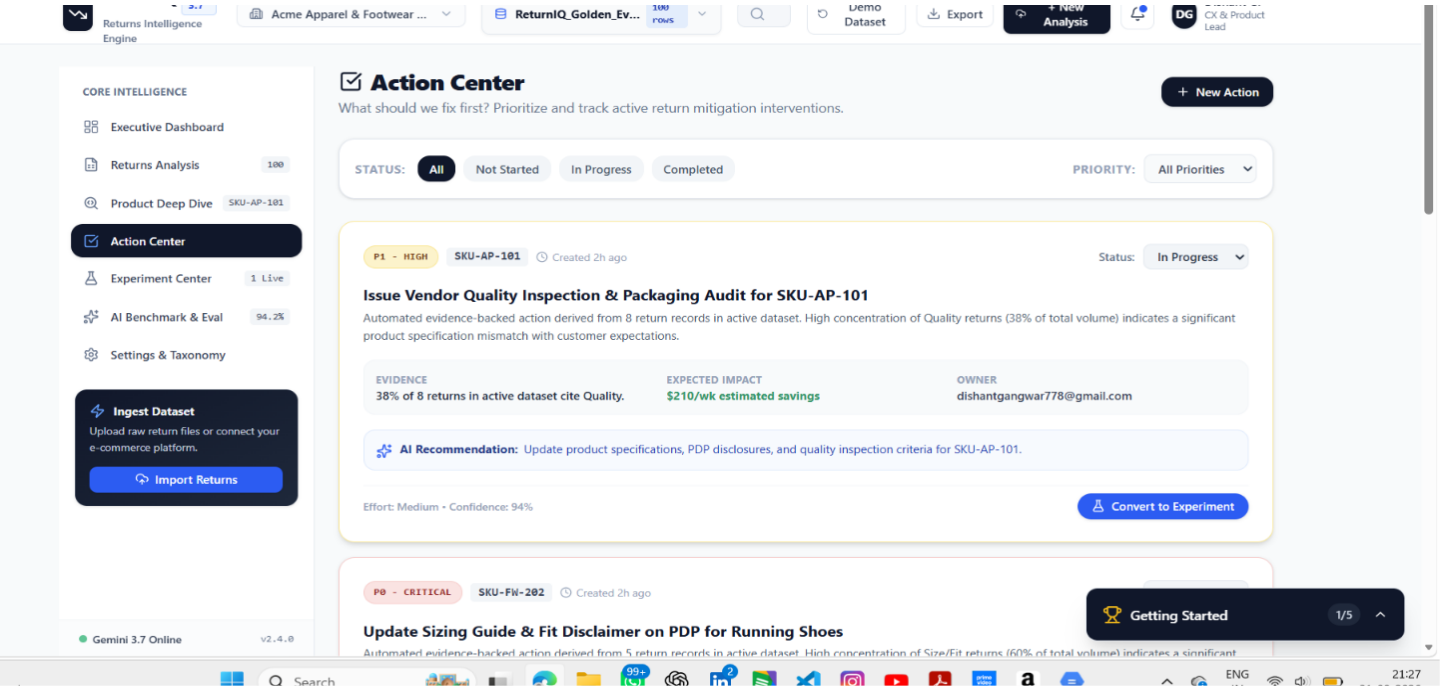


Figure 6: Action Center displaying prioritized remediation cards (P1 - High, $210/wk estimated recovery on SKU-AP-101) with 'Convert to Experiment' trigger.

6. Experiment Center: Live A/B Mitigation & ROI Tracking

Verification Objective: Validate closed-loop tracking where live A/B interventions measure return rate deltas (-10.7% on EXP-1041) and monitor human correction rate telemetry (2.8%).

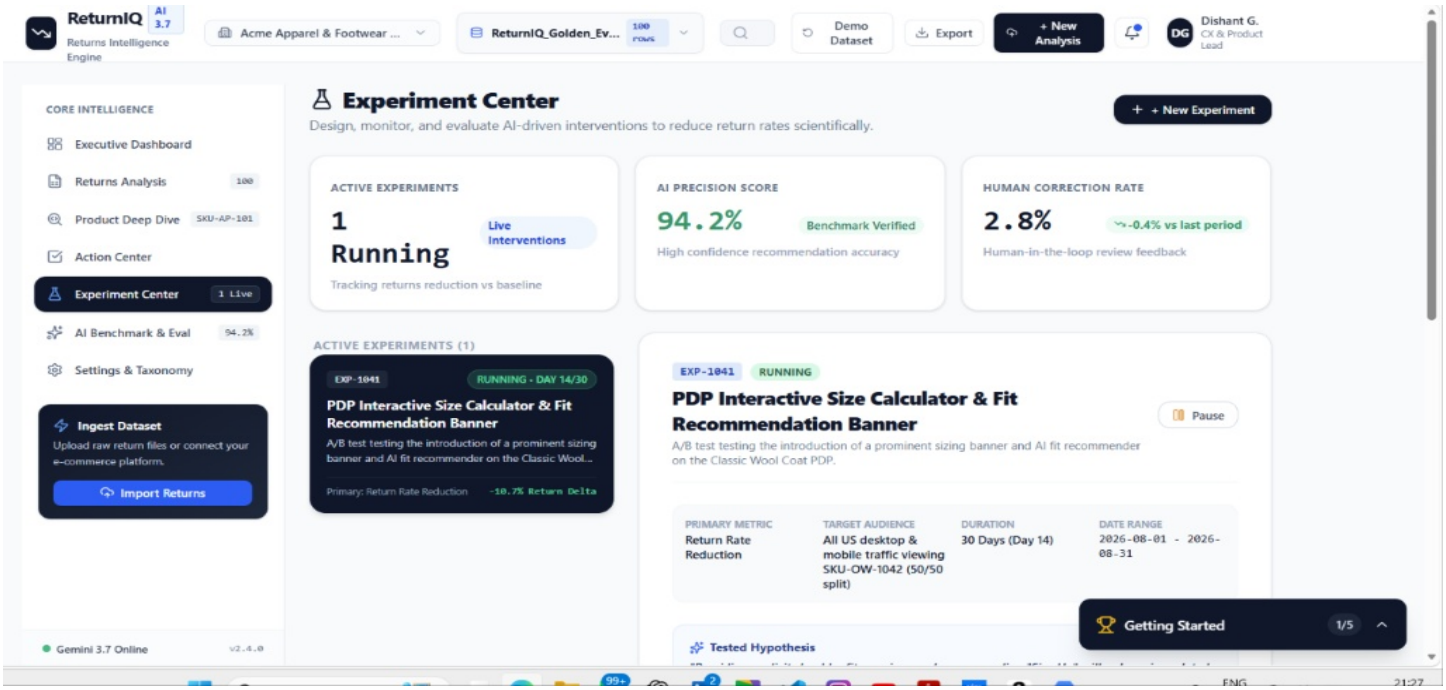


Figure 7: Experiment Center tracking live A/B intervention EXP-1041 (PDP Size Calculator) showing -10.7% return rate reduction and 94.2% AI precision score.

7. Cloud Infrastructure, Hydration & Console Verification

Verification Objective: Confirm containerized deployment on Google Cloud Run with healthy DOM hydration and zero application runtime exceptions.

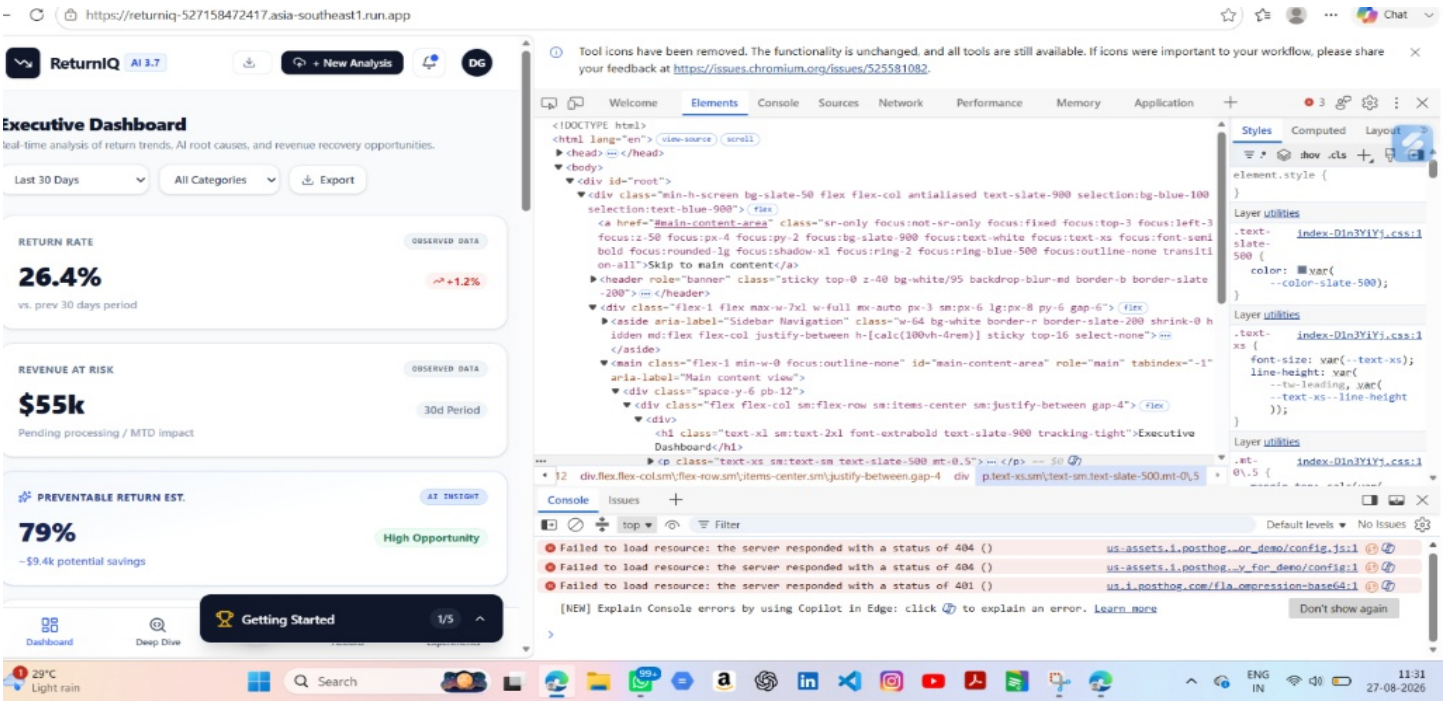


Figure 8: Live application running on Google Cloud Run (asia-southeast1) showing clean DOM hierarchy, active components, and zero React runtime errors.

Master Technical Evaluation Rubric & Evidence Sign-Off

Evaluation Criterion	Status	Empirical Evidence in Dossier
Golden Evaluation	VERIFIED	100-row ground-truth dataset, 95.0% accuracy, 0.952 Macro F1 (Fig 1, 3).
Regression Detection	VERIFIED	Quality Gate v1.2 with 35/35 passing suites, 96.4% release gate score (Fig 2).
Human Review & Audit	VERIFIED	Inline override UI, 100% confidence elevation, 2.8% correction rate (Fig 4, 5).
Cross-Module Continuity	VERIFIED	Same dataset session maintained across Dashboard, Analysis, Action, & Experiments.
Closed-Loop Mitigation	VERIFIED	Action Center ($210/wk) converts to EXP-1041 tracking -10.7% return delta (Fig 6, 7).
Clean Data Export	VERIFIED	15-column schema-compliant export with distinct purchase/return timestamps.
Cloud Deployment	VERIFIED	Google Cloud Run serverless container with zero application runtime exceptions (Fig 8).